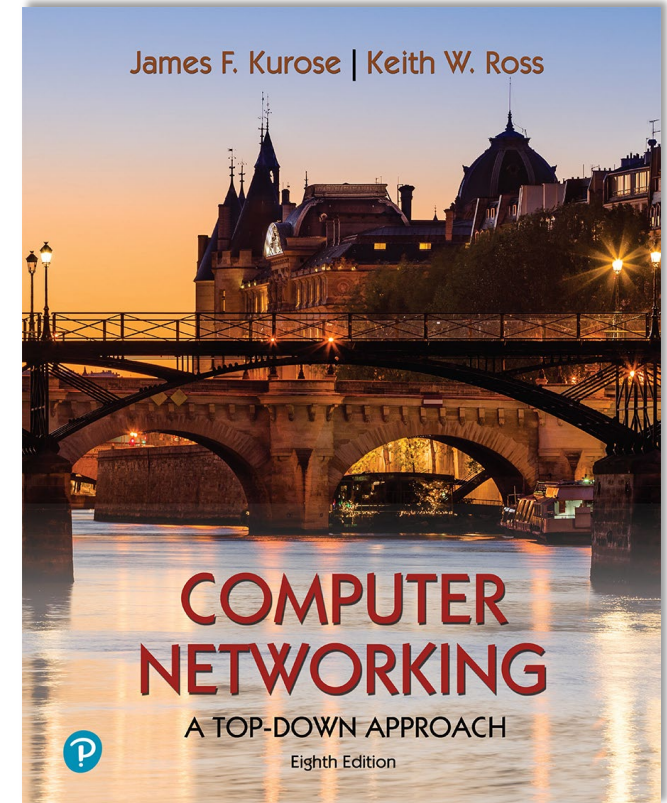


Packet Switched Networks

Acknowledgements

These Slides have been adapted from the originals made available by J. Kurose and K. Ross
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Computer Networking: A Top-Down Approach

8th edition

Jim Kurose, Keith Ross
Pearson, 2020

Goals

- Understanding principles behind packet switched networks
- Introducing some examples of packet switched networks



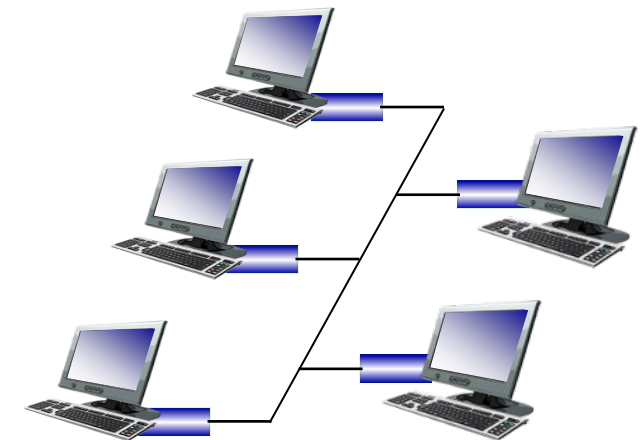
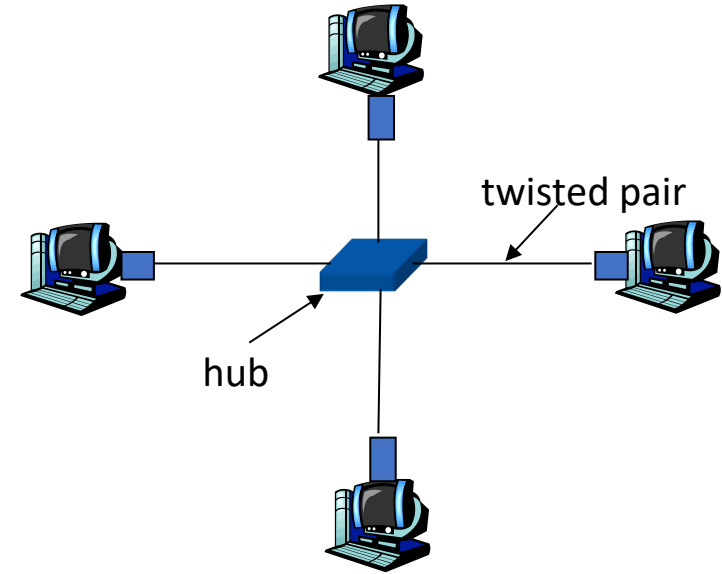
Roadmap

- Link-layer switches
- Switched Ethernet
- Virtual LANs
- Wide-Area Packet Switched Networks
- Link virtualization



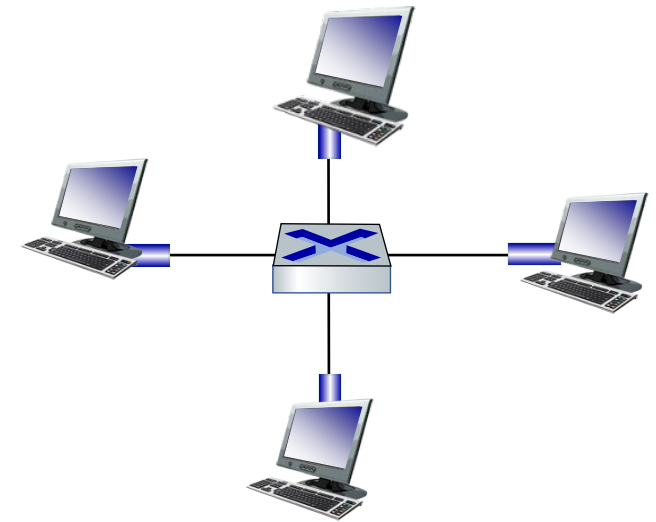
Ethernet Hubs

- Physical-layer (“dumb”) repeater
 - bits coming in one link go out *all* other links at same rate
 - all nodes connected to hub can collide with one another
 - no frame buffering
 - no CSMA/CD at hub: host NICs detect collisions



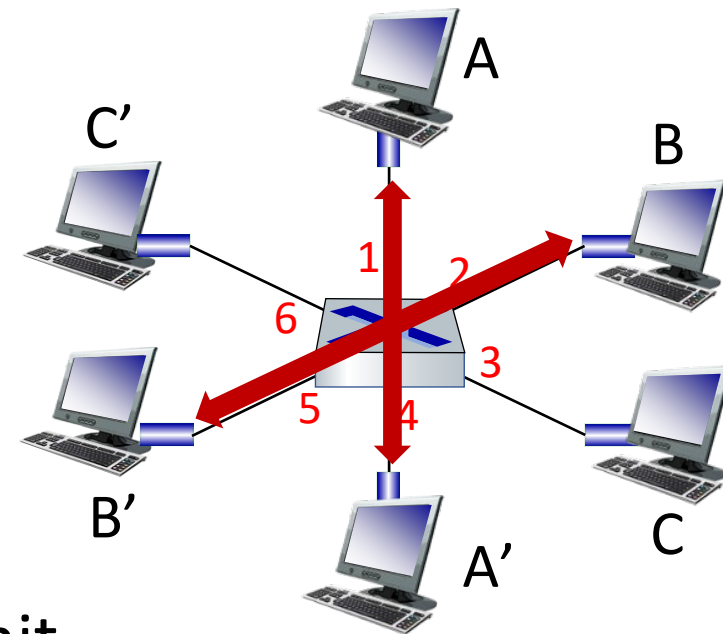
Ethernet switch

- Switch is a **link-layer** device: takes an *active* role
 - store, forward Ethernet frames
 - examine incoming frame's MAC address
 - *selectively* forward frame to one-or-more outgoing links when frame is to be forwarded on segment
 - uses CSMA/CD to access segment
- **transparent**: hosts *unaware* of presence of switches
- **plug-and-play, self-learning**
 - switches do not need to be configured



Switch: multiple simultaneous transmissions

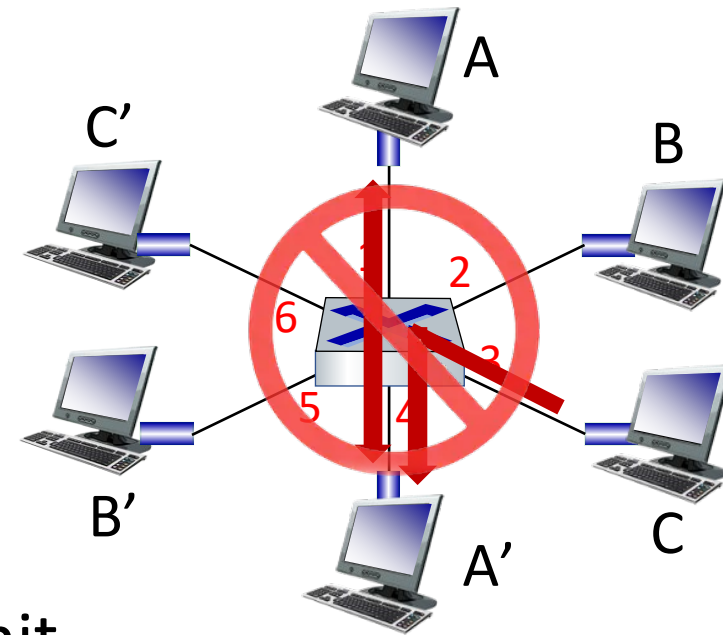
- hosts have dedicated, direct connection to switch
- switches buffer packets
- Ethernet protocol used on *each* incoming link, so:
 - no collisions; full duplex
 - each link is its own collision domain
- **switching:** A-to-A' and B-to-B' can transmit simultaneously, without collisions



switch with six interfaces (1,2,3,4,5,6)

Switch: multiple simultaneous transmissions

- hosts have dedicated, direct connection to switch
- switches buffer packets
- Ethernet protocol used on *each* incoming link, so:
 - no collisions; full duplex
 - each link is its own collision domain
- **switching:** A-to-A' and B-to-B' can transmit simultaneously, without collisions
 - but A-to-A' and C to A' can *not* happen simultaneously



switch with six interfaces (1,2,3,4,5,6)

Switch forwarding table

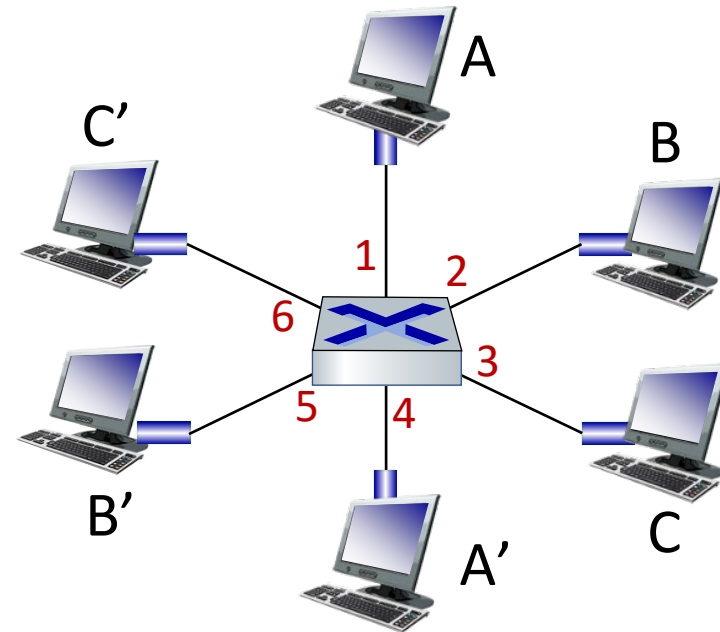
Q: how does switch know A' reachable via interface 4, B' reachable via interface 5?

A: each switch has a **switch table**, each entry:

- (MAC address of host, interface to reach host, time stamp)
- looks like a routing table!

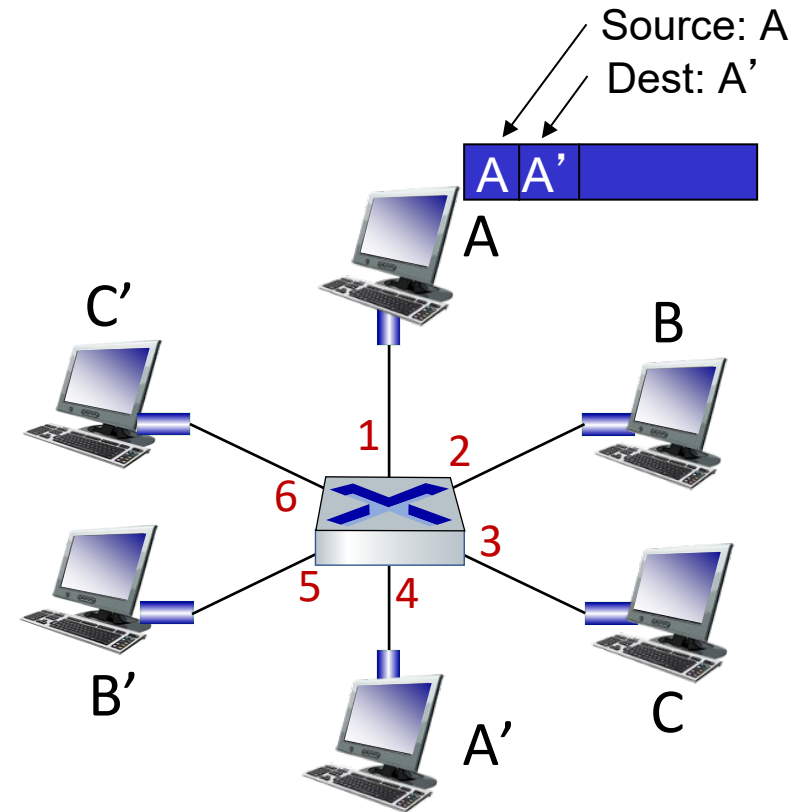
Q: how are entries created, maintained in switch table?

- something like a routing protocol?



Switch: self-learning

- switch *learns* which hosts can be reached through which interfaces
 - when frame received, switch “learns” location of sender: incoming LAN segment
 - records sender/location pair in switch table



MAC addr	interface	TTL
A	1	60

*Switch table
(initially empty)*

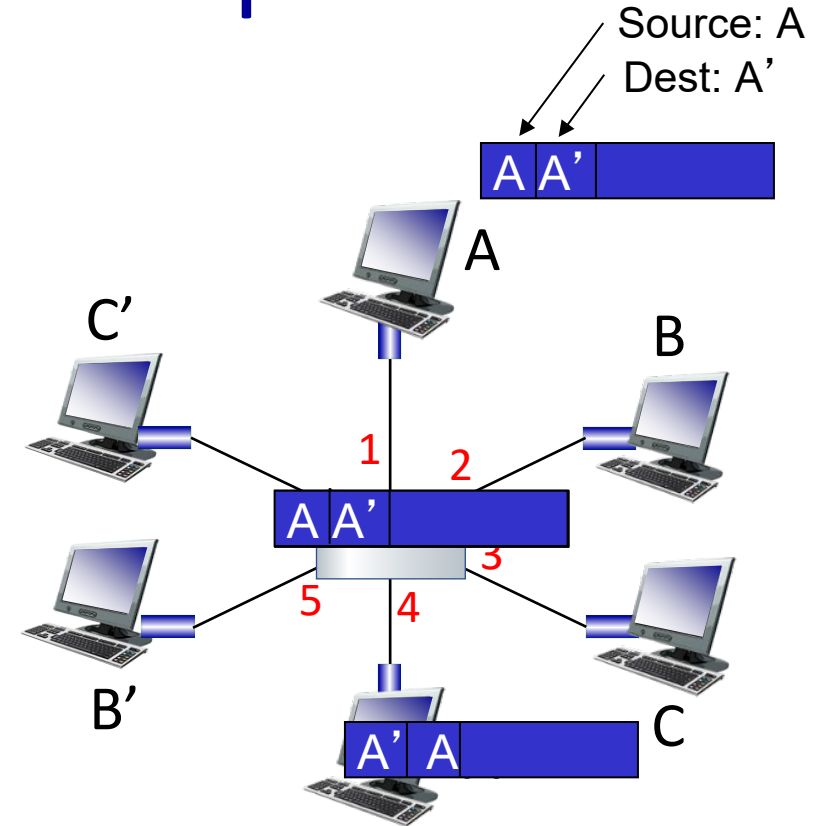
Switch: frame filtering/forwarding

when frame received at switch:

1. record incoming link, MAC address of sending host
2. index switch table using MAC destination address
3. if entry found for destination
 then {
 if destination on segment from which frame arrived
 then drop frame
 else forward frame on interface indicated by entry
 }
 else flood /* forward on all interfaces except arriving interface */

Self-learning, forwarding: example

- frame destination, A',
location unknown: **flood**
- destination A location
known: **selectively send**
on just one link



MAC addr	interface	TTL
A	1	60
A'	4	60

*switch table
(initially empty)*

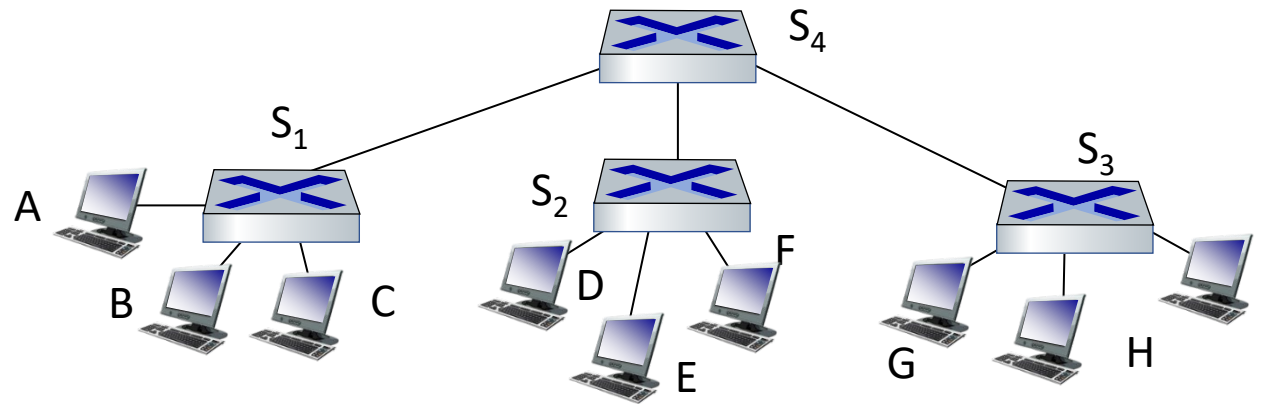
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Interconnecting switches

self-learning switches can be connected together:

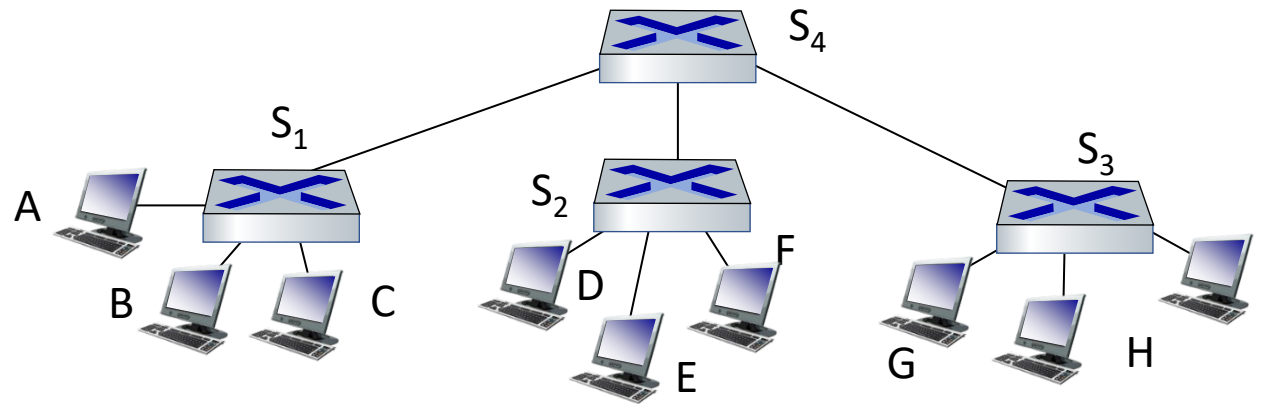


Q: sending from A to G - how does S_1 know to forward frame destined to G via S_4 and S_3 ?

- A: self learning! (works exactly the same as in single-switch case!)

Self-learning multi-switch example

Suppose C sends frame to I, I responds to C



Q: show switch tables and packet forwarding in S_1 , S_2 , S_3 , S_4

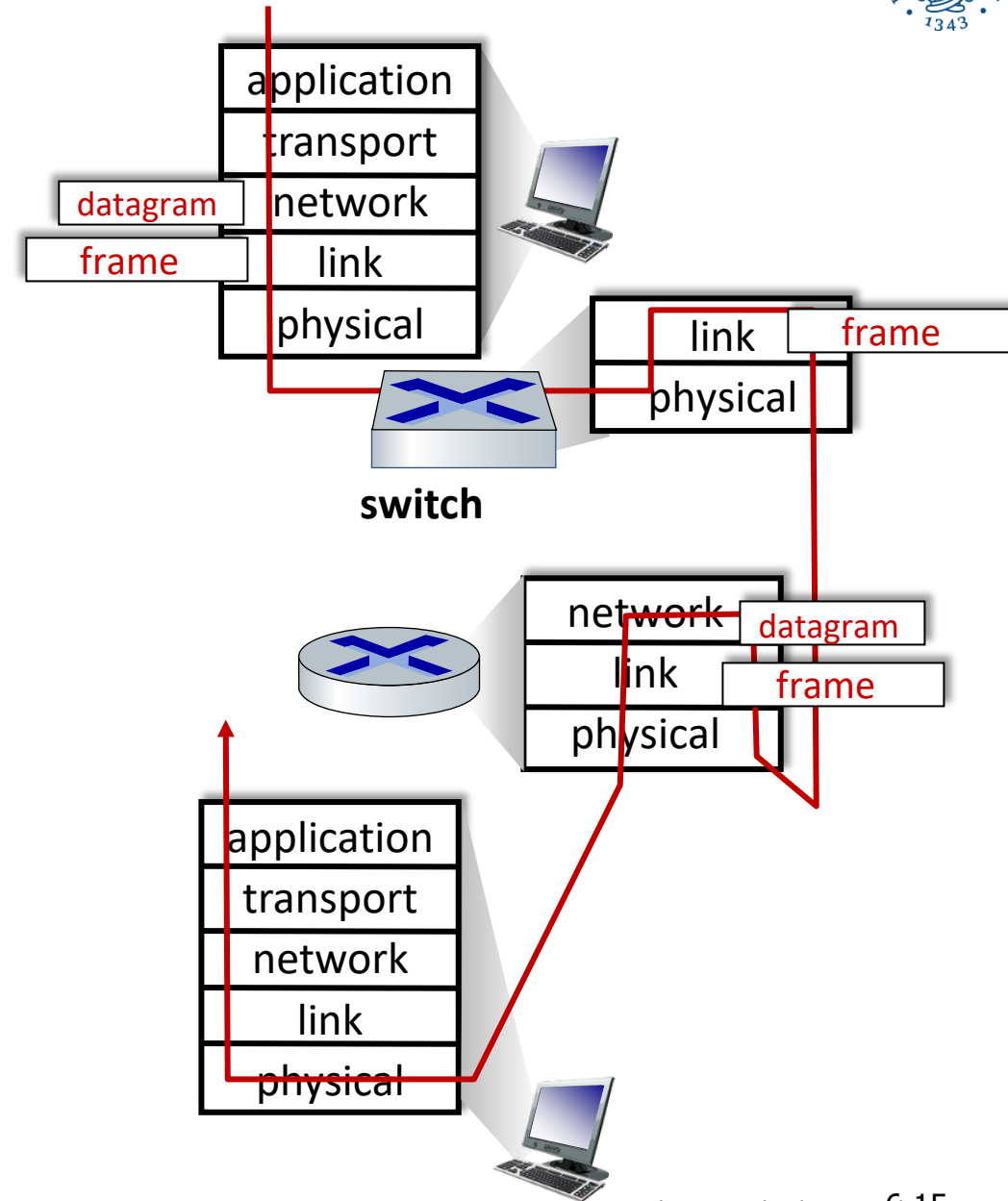
Switches vs. routers

both are store-and-forward:

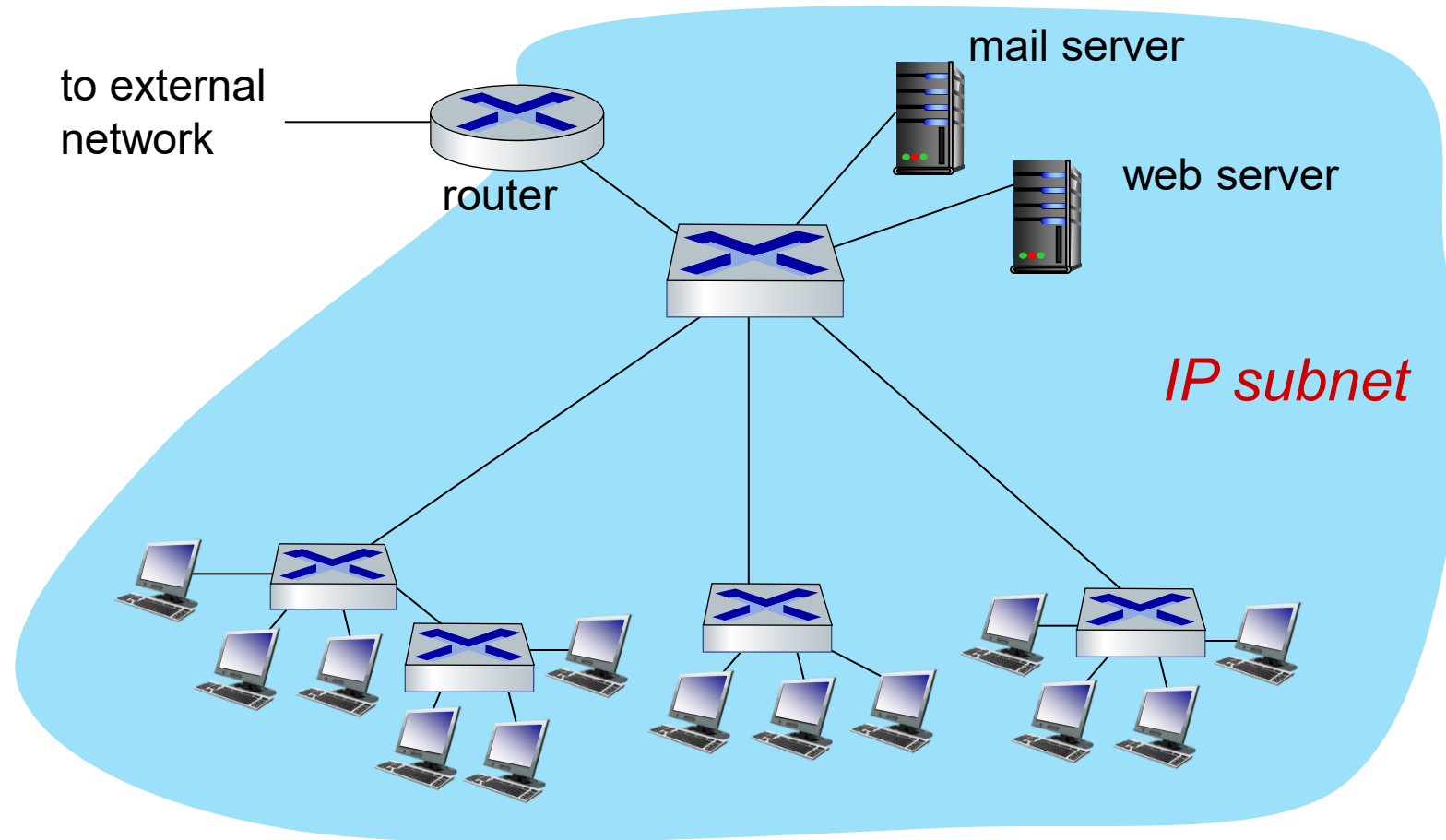
- *routers*: network-layer devices (examine network-layer headers)
- *switches*: link-layer devices (examine link-layer headers)

both have forwarding tables:

- *routers*: compute tables using routing algorithms, IP addresses
- *switches*: learn forwarding table using flooding, learning, MAC addresses



Small institutional network



Datacenter networks

10's to 100's of thousands of hosts, often closely coupled, in close proximity:

- e-business (e.g. Amazon)
- content-servers (e.g., YouTube, Akamai, Apple, Microsoft)
- search engines, data mining (e.g., Google)

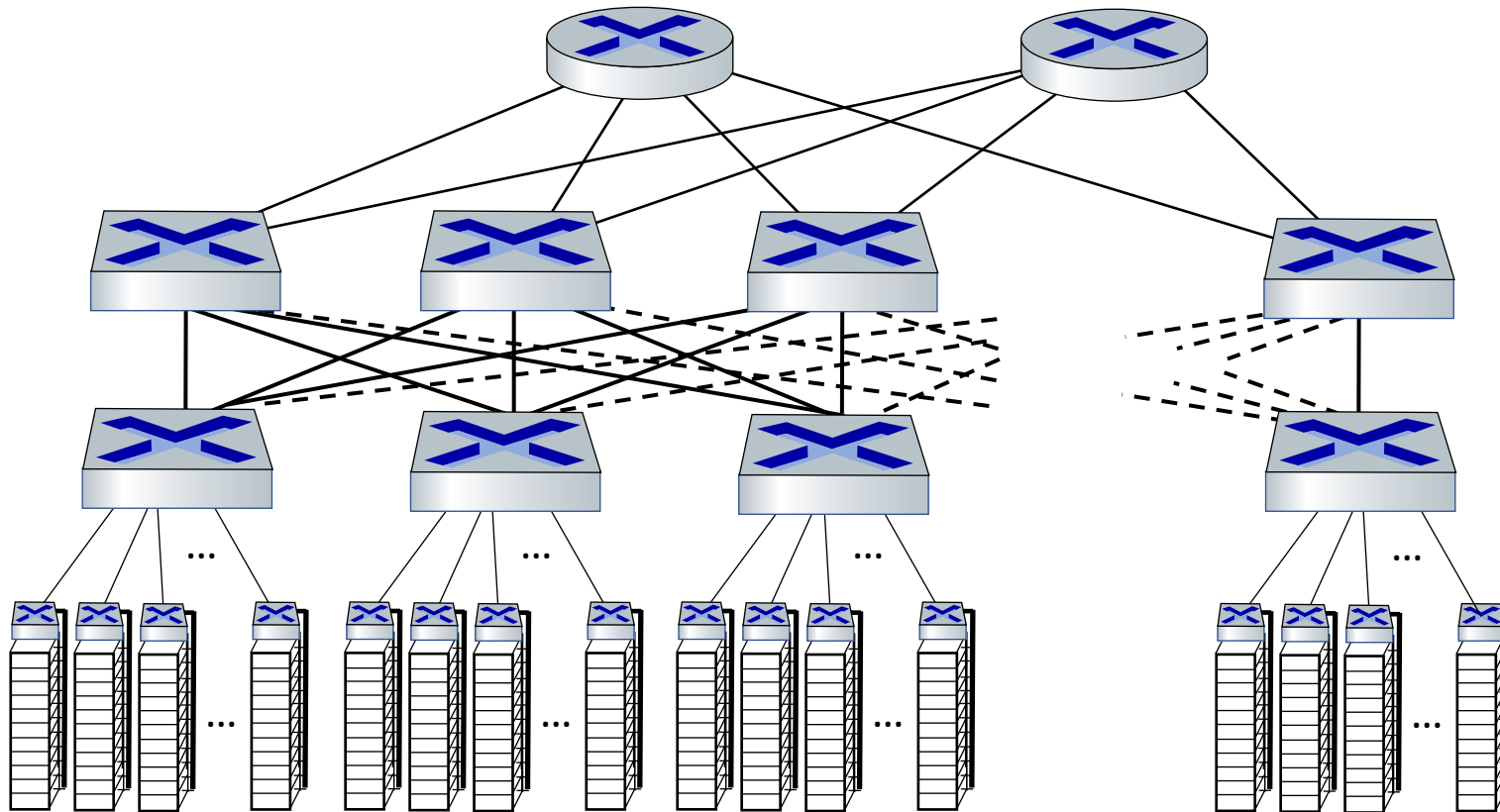
challenges:

- multiple applications, each serving massive numbers of clients
- reliability
- managing/balancing load
- avoiding (processing, networking, data) **bottlenecks**



Inside a 40-ft Microsoft container, Chicago data center

Datacenter networks



Border routers

- connections outside datacenter

Tier-1 switches

- connecting to ~16 T-2s below

Tier-2 switches

- connecting to ~16 TORs below

Top of Rack (TOR) switch

- one per rack
- 40-100Gbps Ethernet to blades

Server racks

- 20- 40 server blades: hosts

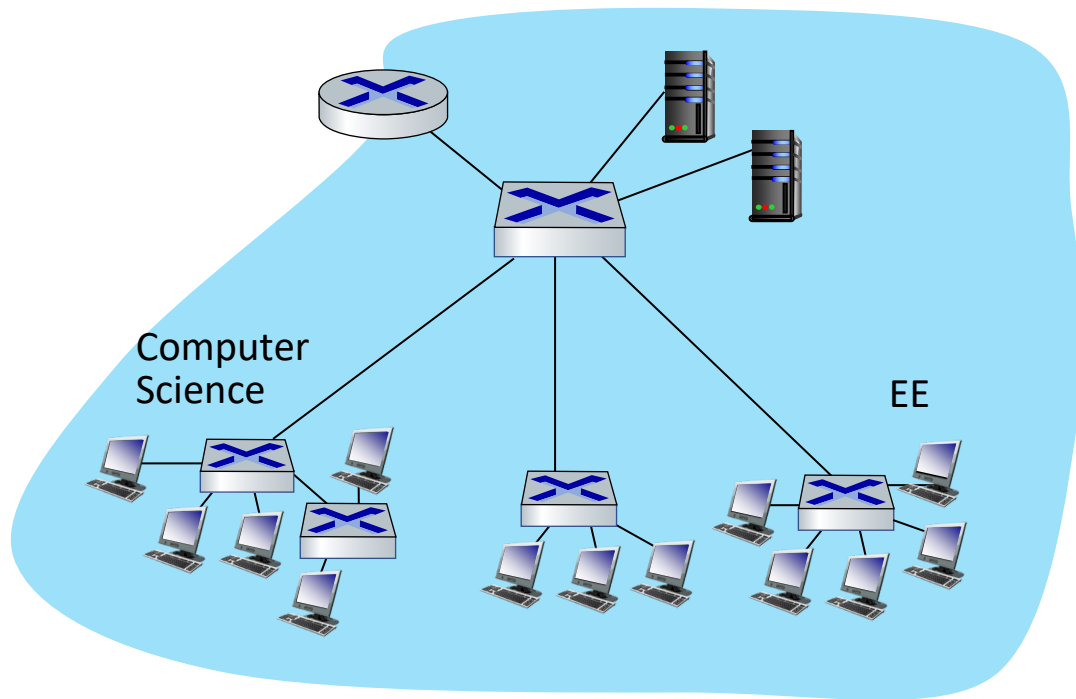
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Virtual LANs (VLANs): motivation

Q: what happens as LAN sizes scale, users change point of attachment?

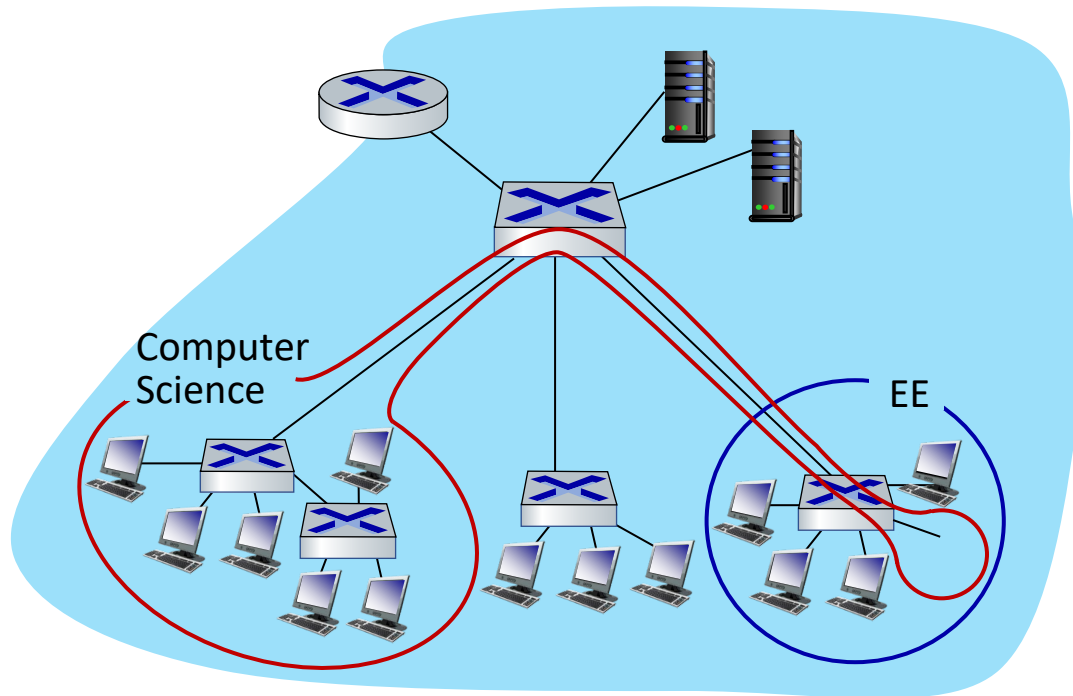


single broadcast domain:

- *scaling*: all layer-2 broadcast traffic (ARP, DHCP, unknown MAC) must cross entire LAN
- efficiency, security, privacy issues

Virtual LANs (VLANs): motivation

Q: what happens as LAN sizes scale, users change point of attachment?



single broadcast domain:

- *scaling*: all layer-2 broadcast traffic (ARP, DHCP, unknown MAC) must cross entire LAN
- efficiency, security, privacy, efficiency issues

administrative issues:

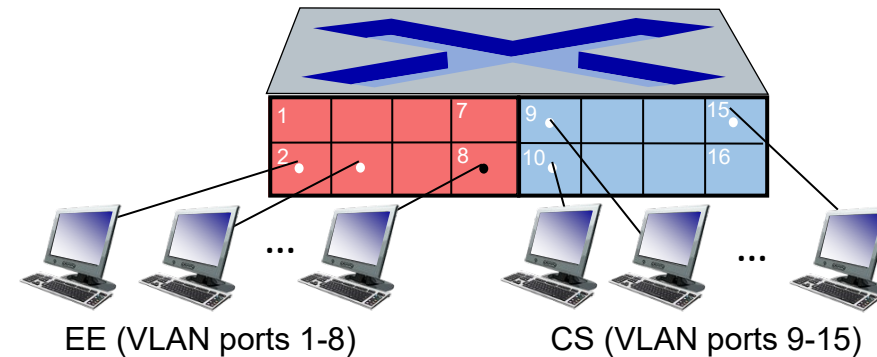
- CS user moves office to EE - *physically* attached to EE switch, but wants to remain *logically* attached to CS switch

Port-based VLANs

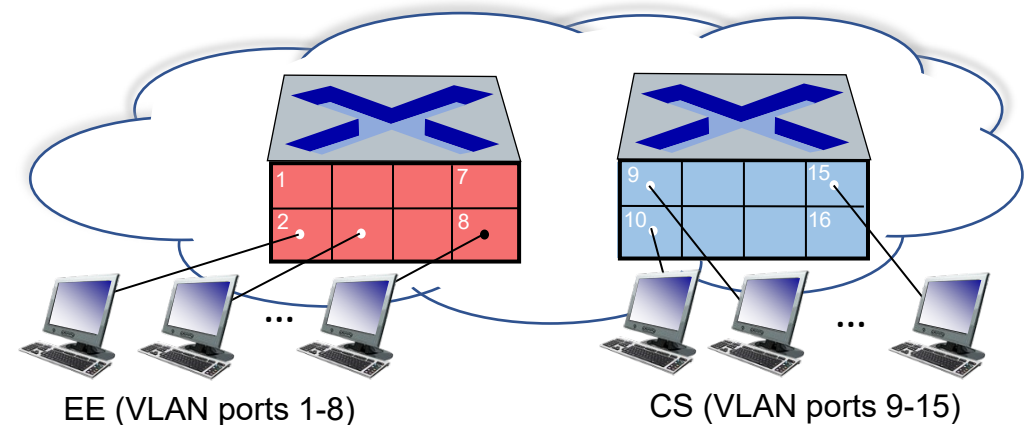
Virtual Local Area Network (VLAN)

switch(es) supporting VLAN capabilities can be configured to define multiple *virtual* LANS over single physical LAN infrastructure.

port-based VLAN: switch ports grouped (by switch management software) so that *single* physical switch

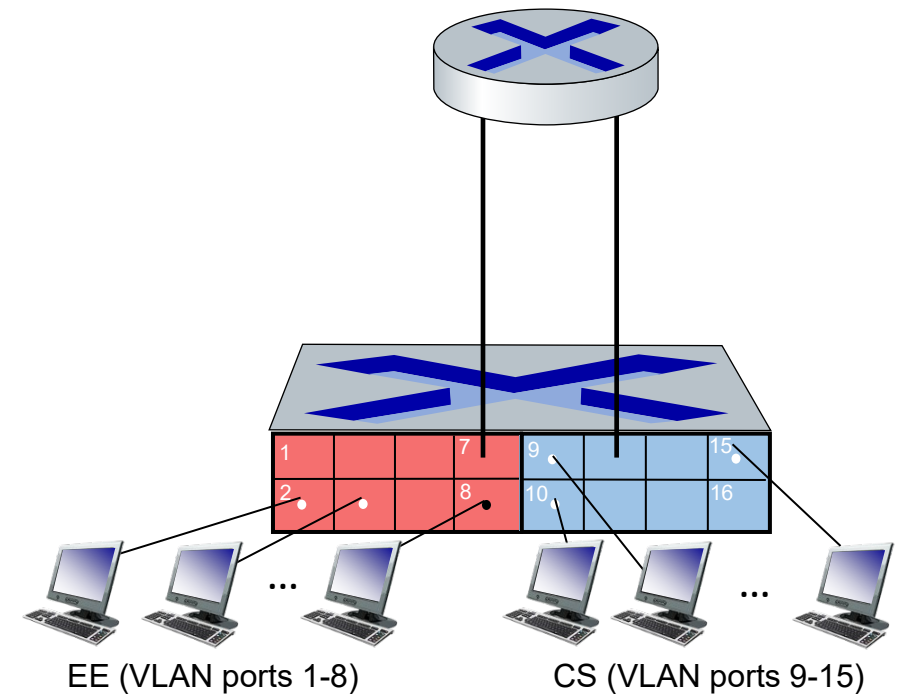


... operates as **multiple** virtual switches

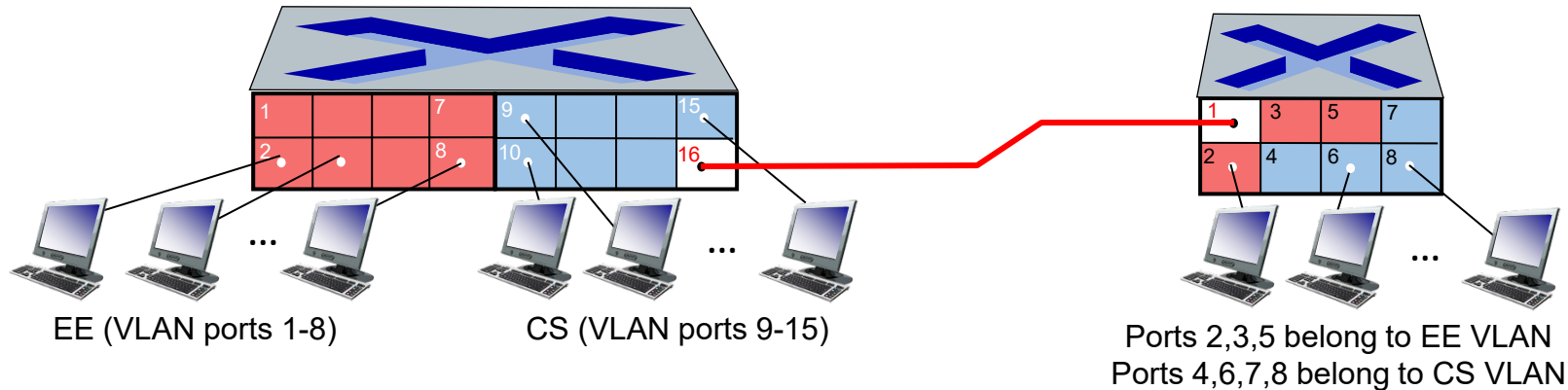


Port-based VLANs

- **traffic isolation:** frames to/from ports 1-8 can *only* reach ports 1-8
 - can also define VLAN based on MAC addresses of endpoints, rather than switch port
- **dynamic membership:** ports can be dynamically assigned among VLANs
- **forwarding between VLANs:** done via routing (just as with separate switches)
 - in practice vendors sell combined switches plus routers



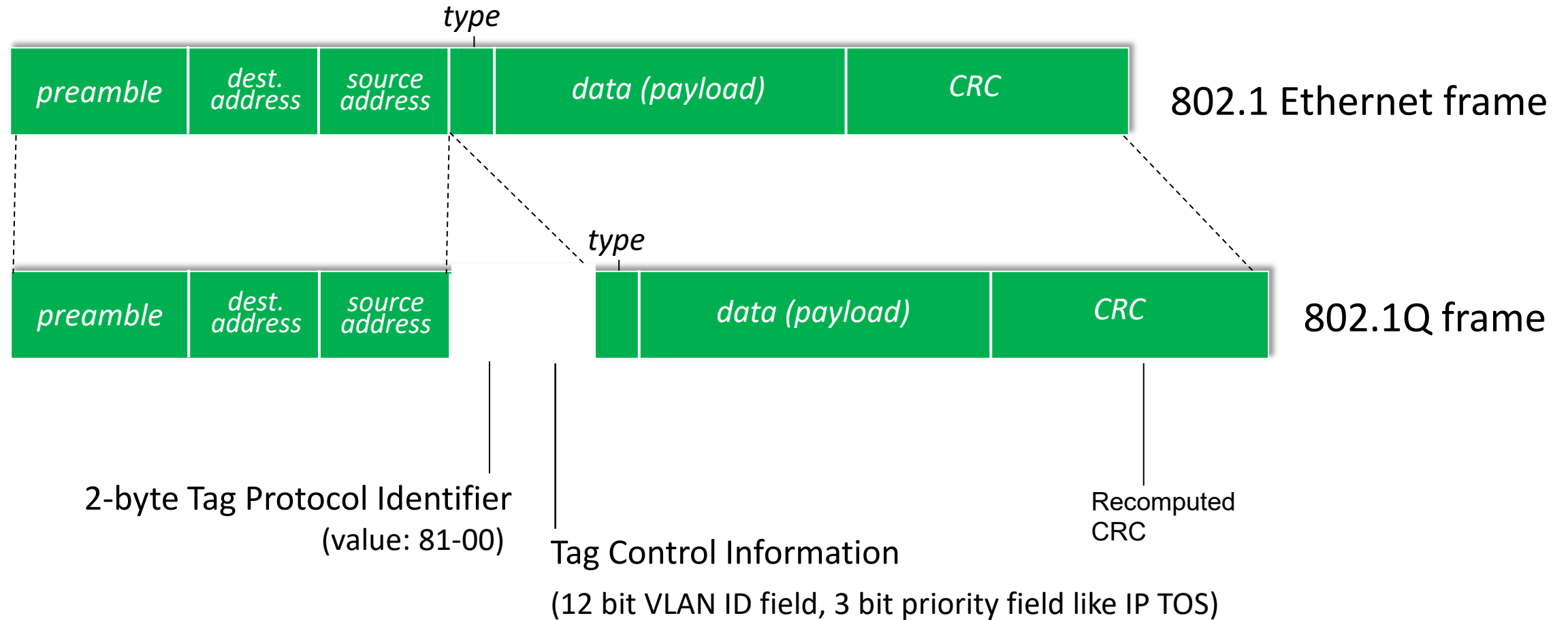
VLANs spanning multiple switches



trunk port: carries frames between VLANs defined over multiple physical switches

- frames forwarded within VLAN between switches can't be vanilla 802.1 frames (must carry VLAN ID info)
- 802.1q protocol adds/removed additional header fields for frames forwarded between trunk ports

802.1Q VLAN frame format

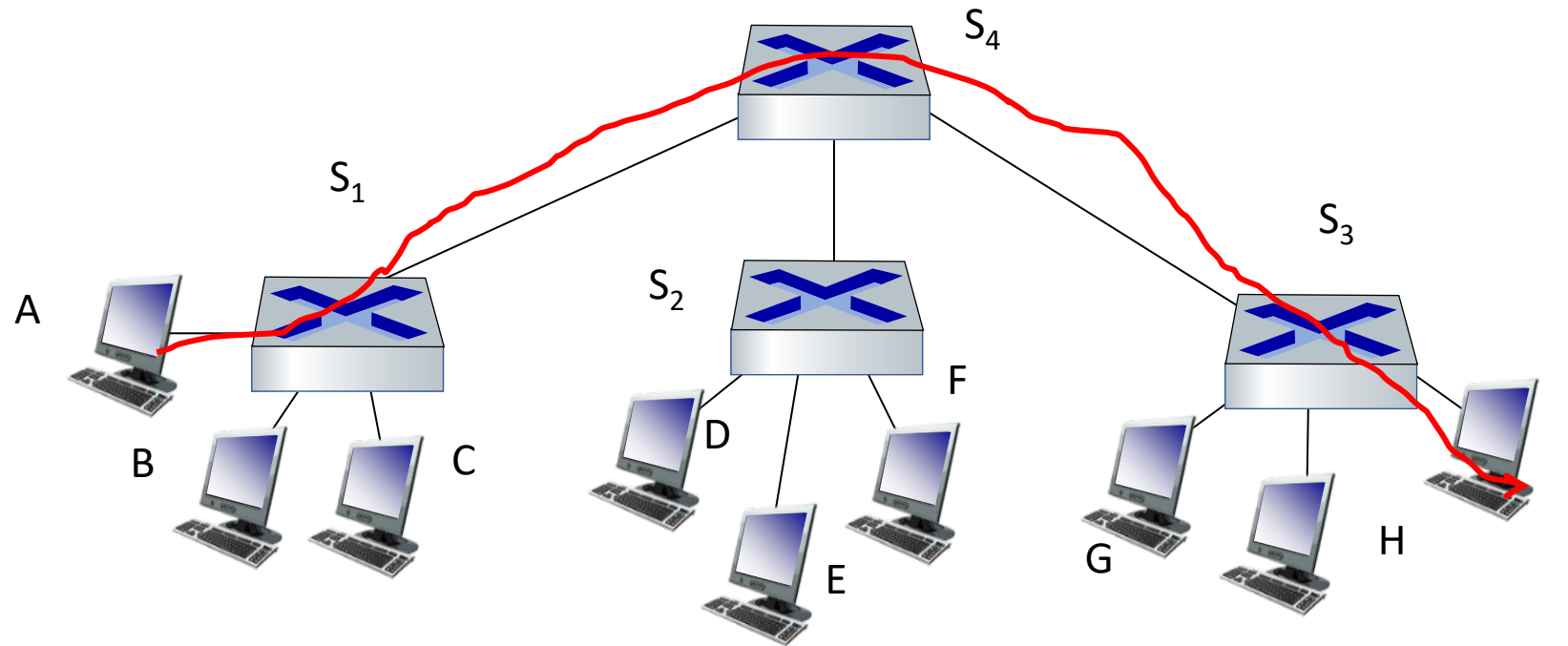


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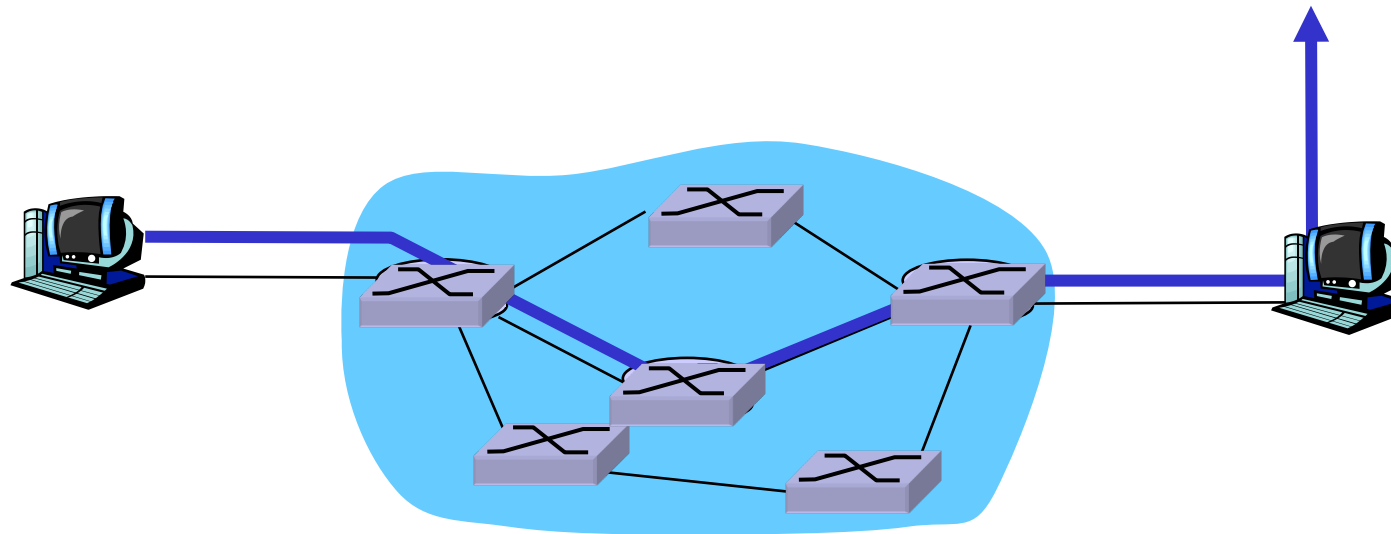


Switched Ethernet



Packet-Switched Wide-Area Networks (WANs)

- Nodes identified through a unique address
 - Similar to the Ethernet MAC address



Type of Service

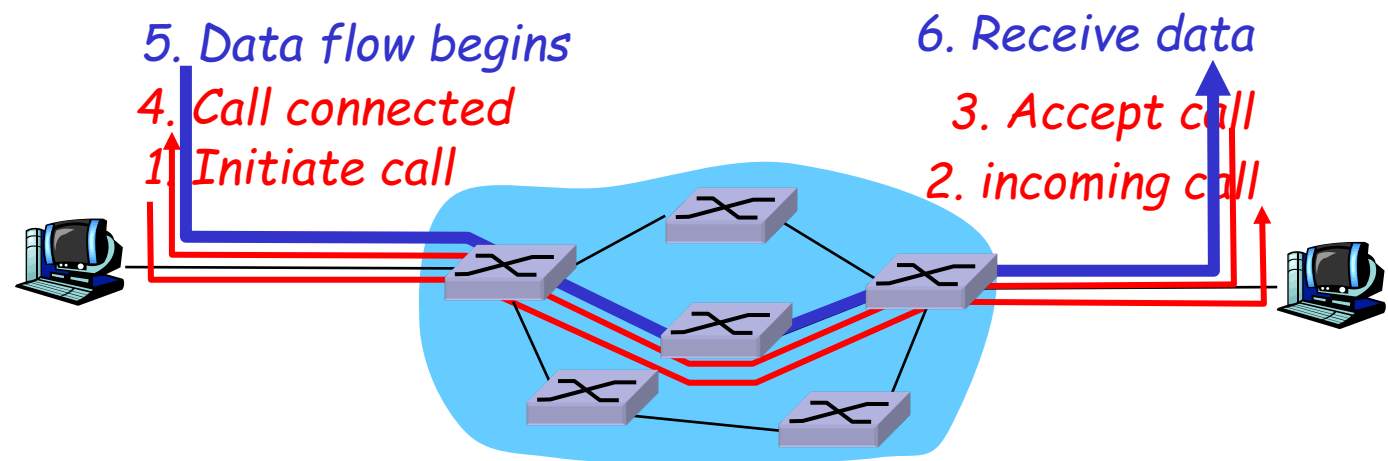
- **Connectionless**
 - Each packet is managed on an individual basis
 - Also known as datagram service
- **Connection**
 - A Virtual Circuit is preliminary established
 - All packets follow the same path

Virtual Circuit

Source-to-destination path

- behaves much like telephone circuit
- performance-wise
- network actions along source-to-dest path

- call setup, teardown for each call *before* data can flow



VC Implementation

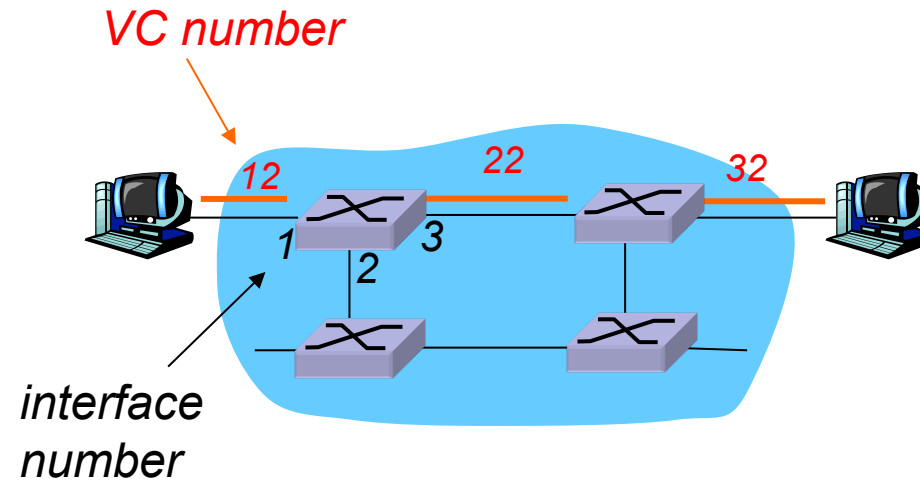
A VC consists of:

1. path from source to destination
2. VC numbers, one number for each link along path
3. entries in forwarding tables in routers along path

- A packet belonging to VC carries VC number
 - rather than dest address
- VC number can be changed on each link
 - New VC number comes from forwarding table

Forwarding Table

Forwarding table in
A switch

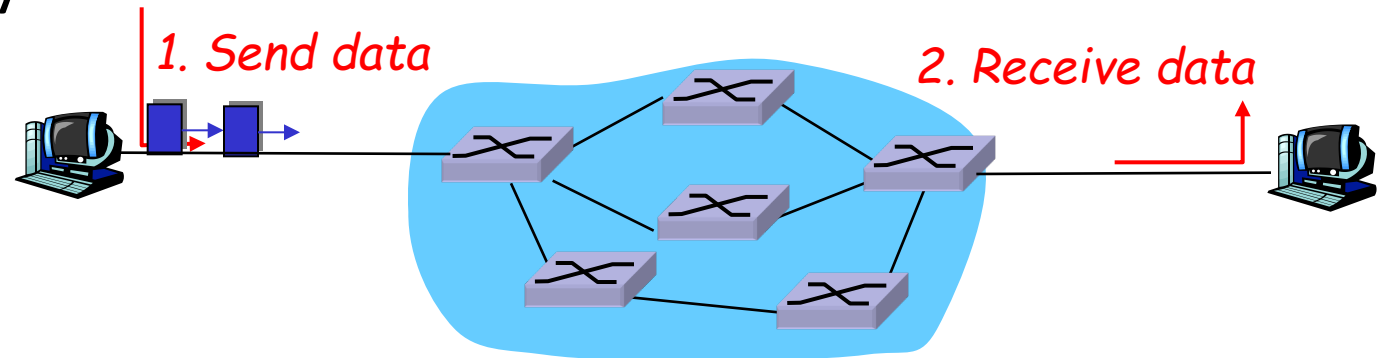


Incoming interface	Incoming VC #	Outgoing interface	Outgoing VC #
1	12	3	22
2	63	1	18
3	7	2	17
1	97	3	87
...	...	...	...

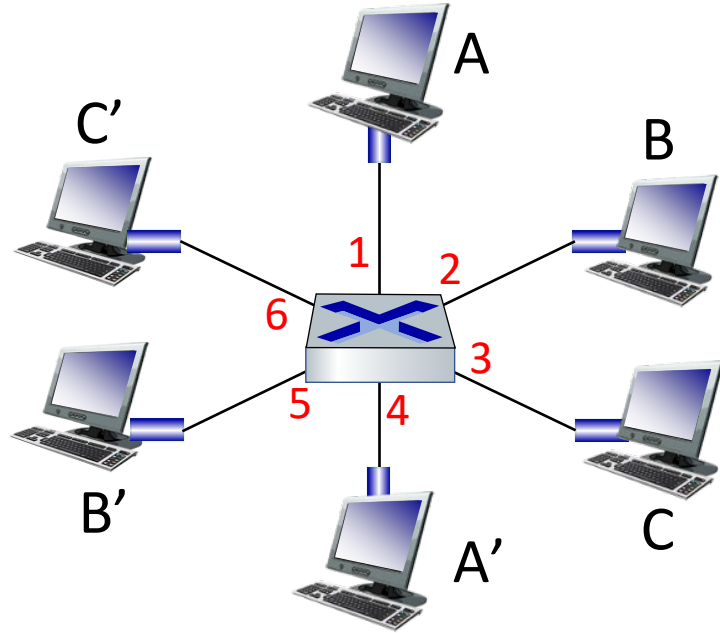
Switches maintain connection state information!

Datagram service

- no call setup
- switches: no state about end-to-end connections
 - no concept of “connection”
- packets between the same source-destination pair may take different paths
- packets forwarded using destination host address



Forwarding Table



MAC address	Interface	TTL
A	1	60
A'	4	60

Forwarding Table

<i>forwarding table</i>	
Destination Address Range	Link Interface
11001000 00010111 00010000 00000000 through 11001000 00010111 00010111 11111111	0
11001000 00010111 00011000 00000000 through 11001000 00010111 00011000 11111111	1
11001000 00010111 00011001 00000000 through 11001000 00010111 00011111 11111111	2
otherwise	3

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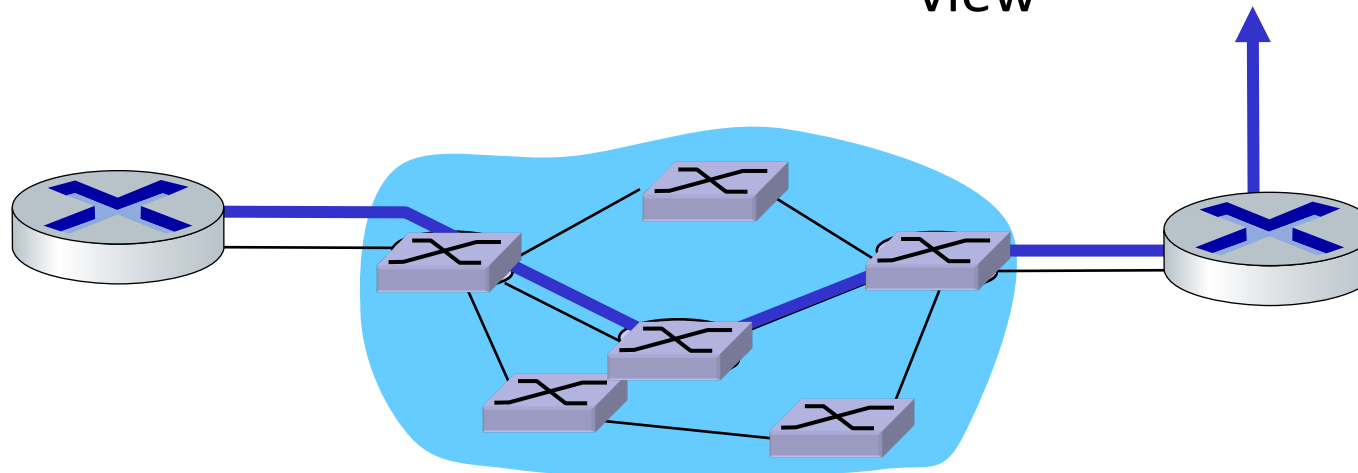
Virtual Links

- Virtualization of resources:
powerful abstraction in systems
engineering:

- virtual memory
- virtual devices
- virtual machines: e.g., java

- Virtual Link:

- The path from S to D is regarded as a point-to-point virtual link
- Just like a physical point-to-point link
- The service type is thus not relevant from the Internet point of view



Summary

- Principles behind packet switched networks
- Switched LANS, VLANs
- Wide-Area Packet-Switched Networks
- Virtualized networks as a point-to point link